GEBZE TEKNİK ÜNİVERSİTESİ

GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep KEPENEK
Öğrenci Numarası Student ID	231040004018
Bölümü Department	Bilgisayar Mühendisliği
Etkinlik Adı Name of the activity	Yapay Zeka Sonuları
Etkinlik Tarihi Date of the activity	19.11.2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	This conference discussed the use of artificial intelligence in architecture and its potential impact on the future of the architectural profession
Özet Summary	This conference focused on how artificial intelligence is used in architecture and whether it will replace architects in the future. AI was presented mainly as a supportive tool in design, analysis and visualization processes. The speaker emphasized that AI does not eliminate creativity but changes the way architects work. Overall, the talk highlighted AI as a transformative technology rather than a direct threat to the profession.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Mimarlık Bölümü
Düzenlendiği Yer Placement	Mimarlık Fakültesi N Blok Turuncu Amfi
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Katılımcı
Katılıma Dair Onay Approval for the participation	Arş.Gör.Sinan Çelik
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep KEPENEK
Öğrenci Numarası Student ID	230104004018
Bölümü Department	Computer Engineering
Etkinlik Adı Name of the activity	Rational Motions of Minimal Quaternionic Degree with Prescribed Plane Trajectories
Etkinlik Tarihi Date of the activity	07/11/2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	This Conference addressed the mathematical structure of rational motions defined using quaternions and how prescribed planar trajectories can be modeled.
Özet Summary	This conference focused on rational motions with minimal quaternionic degree. The speaker explained why quaternions are effective tools for representing rotations and motions in space. It was also shown how motions with prescribed planar trajectories can be constructed using this framework. Overall, the talk was useful for understanding how abstract mathematical concepts are applied to problems in motion and geometry.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Matematik Bölümü
Düzenlendiği Yer Placement	Matematik Bölümü Amfi 2
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Participant
Katılıma Dair Onay Approval for the participation	Doç. Dr. Feray Hacıveliboğlu
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep Kepenek
Öğrenci Numarası Student ID	2301040008
Bölümü Department	Bilgisayar Mühendisliği
Etkinlik Adı Name of the activity	Extremophiles of Fire and Soil: From Türkiye's Geothermal Springs to Stronger, More Nourished Crops
Etkinlik Tarihi Date of the activity	25/11/2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	The seminar introduced microorganism's from Türkiye's geothermal areas and their general importance
Özet Summary	The seminar focused on extremophiles found in Türkiye's geothermal springs and how they survive in high temperature environments. It also noted that Türkiye has many geothermal energy areas. The speaker explained these topics with simple examples and highlighted possible scientific and agricultural uses of such microorganisms such as archaea and bacteria.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Biyoteknoloji
Düzenlendiği Yer Placement	Biyoteknoloji Enstitüsü Amfi (-1. kat)
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Katılımcı
Katılıma Dair Onay Approval for the participation	Biyoteknoloji Enstitüsü Seminer Koordinatörü
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep Kepenek
Öğrenci Numarası Student ID	231044004018
Bölümü Department	Bilgisayar Mühendisliği
Etkinlik Adı Name of the activity	Metallic Materials for Interconnects in Semiconductor Devices
Etkinlik Tarihi Date of the activity	26/11/2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	This conference focused on metallic interconnect materials used in semiconductor devices, their properties, challenges, and next-generation material solutions.
Özet Summary	This conference examined metallic interconnect materials used in semiconductor devices. As device dimensions scale down, issues such as increased resistance, electromigration and reliability problems in traditional materials like copper were discussed. Possible solutions, including new metal alloys and alternative materials, were also presented. The talk was useful for understanding the role of nanotechnology and material science in modern semiconductor technologies.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Nanoteknoloji Enstitüsü
Düzenlendiği Yer Placement	GTU A Blok Konferans Salonu
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Katılımcı
Katılıma Dair Onay Approval for the participation	
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep Kepenek
Öğrenci Numarası Student ID	231104004018
Bölümü Department	Bilgisayar Müh.
Etkinlik Adı Name of the activity	Mutual visibility Problems in Graphs
Etkinlik Tarihi Date of the activity	05/12/2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	The event focused on mutual visibility problems, which study when vertices in graph can see each other.
Özet Summary	In this seminar, we learned what mutual visibility means in graph theory. The speaker explained that two vertices are visible to each other when no other vertex or path blocks the connection between them. It was interesting to see how this problem changes in different types of graphs. The talk also mentioned applications in robotics and sensor networks. Overall, it was a simple but informative presentation.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Matematik Bölümü
Düzenlendiği Yer Placement	Matematik Bölümü Amfi 2
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Katılımcı
Katılıma Dair Onay Approval for the participation	Doç. Dr. Gülden GÜN POLAT GTÜ Matematik Bölümü
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Raporu GTU 110 Scientific and Technological Activities Report	
Öğrencinin Adı-Soyadı Name and Surname	Recep Keperek
Öğrenci Numarası Student ID	231044044018
Bölümü Department	Bilgisayar Müh.
Etkinlik Adı Name of the activity	Defective Ramsey Numbers: Classical Proofs and Computer Enumerations
Etkinlik Tarihi Date of the activity	28/11/2025
Etkinliğin Konusu/İçeriği Subject/Content of the Activity	the talk focused on classical proofs and computer based enumeration of defective Ramsey numbers.
Özet Summary	In this conference, the topic of Defective Ramsey Numbers in graph theory was discussed. While classical Ramsey Numbers guarantee the existence of certain subgraph structures, defective Ramsey Numbers allow a limited number of missing edges in these structures. The speaker first briefly reviewed the basic definition of Ramsey numbers and classical proof techniques, then introduced the defective case. The use of computer based enumeration methods to obtain bounds for these numbers was also explained. Overall, the talk was useful in showing the intersection between theoretical graph theory and computational methods.
Düzenleyen Bölüm/Akademik Birim Department/Academic Unit	Matematik
Düzenlendiği Yer Placement	Matematik bölümü Amfi 2
Öğrencinin Etkinlikteki Rolü (katılımcı/düzenleyici, vb.) Student's Role in the Activity (participant/organizer, etc.)	Katılımcı
Katılıma Dair Onay Approval for the participation	Dr. Ar. Gör. Fatih YETGİN
Öğrencinin imzası Student's Signature	

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GTU 110 Bilimsel ve Teknolojik Etkinlik Dersi Raporu GTU 110 Scientific and Technological Activities Course Report	
Öğrencinin Adı Soyadı Name and Surname	Recep KEPENEK
Öğrenci Numarası Student ID	230104004018
Bölümü Department	Computer Engineering
Kazanımlar Outcomes	This semester, the GTU 110 course gave me a unique opportunity to step outside the computer lab and see how science applies to the real world. As a Computer Engineering student, I often focus on code and algorithms, but attending six different conferences—ranging from architecture to abstract mathematics and biotechnology—broadened my perspective significantly. First, the conference "AI Questions" was particularly relevant to my field. It focused on how AI is used in architecture not to replace humans, but as a supportive tool for design and analysis. Learning that AI transforms creativity rather than eliminating it gave me a more ethical and human-centric view of the software we build. A significant portion of my activities focused on the mathematical foundations of computer science. In the "Mutual Visibility Problems in Graphs" seminar, I learned how graph theory models visibility between vertices, which has direct applications in robotics and sensor networks. Similarly, the talk on "Defective Ramsey Numbers" demonstrated how computer-based enumerations are used to solve complex theoretical proofs, showing the bridge between pure math and computational power. Additionally, the conference on "Rational Motions" helped me understand quaternions, which are essential for modeling 3D rotations in computer graphics and game development. On the hardware and science side, the "Metallic Materials for Interconnects" seminar was crucial for understanding the physical limits of the processors we program. I learned about the reliability issues in semiconductor devices as they shrink, reminding me that software performance is deeply tied to material science. Finally, the "Extremophiles of Fire and Salt" talk introduced me to biotechnology, showing how microorganisms in Turkey's geothermal springs can be analyzed for agricultural uses. In conclusion, these activities taught me that engineering is not just about isolated technical skills. Whether it's the math behind a robot's vision or the materials inside a chip, I realized that an interdisciplinary approach is essential for solving modern problems.